

GreenMach Sustainability Portfolio

Team SilverMach · STEM Racing Open Nationals 2026 — environmental, social and economic sustainability engineered alongside the race car itself.

What Is GreenMach?

GreenMach is Team SilverMach's dedicated sustainability programme — the engineering decisions, material choices and community commitments that sit alongside our race car, not apart from it. Every gram printed, every CO₂ cartridge fired and every workshop run is logged against the same standard we hold our lap times to: minimise material, maximise reuse, and make sure the knowledge we build reaches more students than just our five-person team.

Three Pillars of Sustainability

Environmental

A 35–45g printed assembly, bio-based PLA and offcut/cartridge recycling keep material use to the minimum the design allows.

Social

150+ students reached through 3 community partnerships bringing CAD, physics and race data into local classrooms.

Economic

In-house manufacturing, reusable tooling and transparent sponsor reporting keep costs low and impact measurable.

Five UN Sustainable Development Goals

04

Quality
Education

08

Decent Work &
Econ. Growth

11

Sustainable Cities
& Communities

12

Responsible
Consumption

13

Climate
Action

Impact in Numbers

150+

STUDENTS REACHED

3

COMMUNITY PARTNERS

5

SDGs SUPPORTED

35–45g

FINAL PRINTED ASSEMBLY

Programme Timeline

Q1 2026

Programme Defined

Scoped as a formal sustainability track alongside the Open Nationals build.

Q2–Q3 2026

150+ Students Reached

Outreach crossed the 150-student mark across all partner sites.

Q1–Q2 2026

Material & Process Audit

Every part reviewed for material, manufacturing method and reuse potential.

9 Aug 2026

STEM Racing Open Nationals

The 35–45g GreenMach-audited assembly races at Open Nationals 2026.

Q2 2026

First Community Workshop

First of three partnerships launched, bringing CAD and physics demos to class.

Post-Season

Portfolio & Knowledge Handover

Portfolio published and shared with sponsors, partners and the next cohort.